Exp: 6 Write the python program for Vacuum Cleaner Problem.

Aim:

To write a python program for vacuum cleaner problem.

Algorithm:

Step 1: Create a data structure to represent the state of the problem, including the current position of the vacuum cleaner and the state of each square in the room (clean or dirty).
Step 2: Create a data structure to represent the state space of the problem, including the initial state, the goal state, and the possible transitions between states.
Step 3: Use a search algorithm, such as depth-first search or breadth-first search, to explore the state space and find a solution.
Step 4: At each step of the search, generate all possible successor states from the current state by applying one of the allowed actions (move up, down, left, right, or clean).
Step 5: Check if the successor state is valid (i.e., the vacuum cleaner cannot move outside the room, and it can only clean dirty squares).
Step 6: If the successor state is valid, add it to the search tree and continue the search.
Step 7: If the successor state is the goal state, return the path from the initial state to the goal state.
Step 8: If there are no more possible successor states to explore, backtrack to the previous state and try another possible action.

Program:

```
import random

room = [[random.randint(0,1) for j in range(4)] for i in range(4)]

print("All the rooms are dirty")

print([[1,1,1,1],[1,1,1,1],[1,1,1,1],[1,1,1,1]])

print("Before cleaning the room I detect all of these random dirts")

print(room)

cleaned = 0

total = 16

for i in range(4):

    for j in range(4):

        if room[i][j] == 1:
```

```
            print("Vacuum in this location now,", i, j)
            room[i][j] = 0
            cleaned += 1
            print("cleaned", i, j)
print("Room is clean now, Thanks for using : A.SAFARJI CLEANER")
print(room)
performance = (cleaned / total) * 100
print("performance =", performance, "%")
```

Output :

```
>>>
All the rooms are dirty
[[1, 1, 1, 1], [1, 1, 1, 1], [1, 1, 1, 1], [1, 1, 1, 1]]
Before cleaning the room I detect all of these random dirts
[[0, 1, 1, 1], [0, 1, 1, 1], [1, 1, 1, 0], [0, 0, 1, 0]]
Vacuum in this location now, 0 1
cleaned 0 1
Vacuum in this location now, 0 2
cleaned 0 2
Vacuum in this location now, 0 3
cleaned 0 3
Vacuum in this location now, 1 1
cleaned 1 1
Vacuum in this location now, 1 2
cleaned 1 2
Vacuum in this location now, 1 3
cleaned 1 3
Vacuum in this location now, 2 0
cleaned 2 0
Vacuum in this location now, 2 1
cleaned 2 1
Vacuum in this location now, 2 2
cleaned 2 2
Vacuum in this location now, 3 2
cleaned 3 2
Room is clean now, Thanks for using : A.SAFARJI CLEANER
[[0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0]]
performance = 62.5 %
>>>
```

Result:

Hence, the simple python program for Vacuum Cleaner Problem was done .